

MTH205 Analysis in one variable

Tutorial 3

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- (1) Show that the function $f(x) = |x|$ for all $x \in \mathbb{R}$ is continuous on $\mathbb{R}$.
- (2) Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous such that $f(r) = 0$ for every $r \in \mathbb{Q}$ then prove that $f(x) = 0, \forall x \in \mathbb{R}$.
- (3) If $x \in \mathbb{R}$ we define $\lfloor x \rfloor$ to be the greatest integer $n \in \mathbb{Z}$ with $n \leq x$. Determine the points of continuity of the following functions defined on $\mathbb{R}$:
- (a) $f(x) = \lfloor x \rfloor$ (c) $h(x) = \lfloor \sin(x) \rfloor$
- (b) $g(x) = x \lfloor x \rfloor$ (d) $\varphi(x) = \lfloor \frac{1}{x} \rfloor$ for $x \neq 0$.
- (4) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Then prove that $S = \{x \in \mathbb{R} / f(x) = 0\}$ is a closed subset of $\mathbb{R}$.
- (5) Prove that the function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x) = \begin{cases} 0, & x = 0 \\ x \sin(\frac{1}{x}), & x \neq 0 \end{cases}$$

is continuous at $x = 0$.